COLLECTION OF CTL DECISIONS

Standard:	**Sub clause:**	**Sheet n. 371**
IEC 61010-1:1990 +A1:1992 + A2 :1995	6.5.2	**Page 1(1)**
Subject: Double insulation and reinforced insulation	**Key words:** - Vacuum encapsulated transformer	**Decision taken by ETF3 and confirmed by CTL at its 38th meeting, in Toronto**

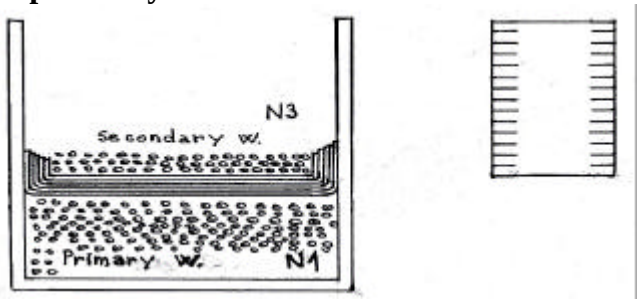
Question:

Can we accept that the insulation between primary and secondary windings of a Vacuum Encapsulated Transformer comply with requirements of double insulation defined in sub-claue 6.5.2.

Background

There is a conventional transformer with primary and secondary windings 220Vac / 12 Vac. Both windings are wound on the same part of the bobbin and separated by 8 layers of insulation tape. The thickness of the tape is 0,05 mm which gives the distance through insulation between primary and secondary windings close to the end of the bobbin at least 0,4 mm. . The insulation tape has fringes and is depicted in the attached drawing. The transformer is vacuum impregnated by AralditCW2245N.
The transformer has passed the test of . 2.9.6 and 2.9.7 of IEC60950 with humidity testing performed at 40C. This means that tests performed (temperature cycling test, Hipot test with 3000x1,6Vac and sectioning after temperature cycling test) showed that there are no voids nor cracks in the compund.

Decision:

These measures taken are adequate to satisfy the needs of IEC61010-1

Explanatory Notes: